

Innovations in semiconductor materials, such as gallium nitride and silicon carbide, offer smaller chip sizes and lower emissions per chip.

Searching for sustainable solutions

IC manufacturers are focusing on developing new etching processes and photoresist materials. A key research area involves polyfluoroalkyl substances (PFAS), known for their robust carbon-fluorine bonds and resistance to atmospheric degradation. While PFAS are effective in semiconductor production, they pose environmental risks due to their persistence and potential harm to wildlife and human health at high exposure levels.

The search for safer PFAS alternatives is ongoing, with the industry aiming to phase them out in applications such as PFAS photoresists by 2030. Notable successes include replacing hazardous substances like N-methylpyrrolidone in photolithography processes. Major chemical manufacturers introduced safer NMP-free solvents in 2021, and Taiwan Semiconductor Manufacturing Company (TSMC) adopted them in 2023.

Innovative strategies also involve modifying the molecular structure of etching chemicals. For instance, replacing tetrafluoromethane (CF₄) with CF₃I, which substitutes an iodine atom for a fluorine atom, weakens the strong carbon-fluorine bond, reducing the chemical's environmental impact. Lam Research and Tokyo Electron are investing in these sustainable alternatives for dry etching and plasma processes.

These advancements are promising, but it remains to be seen whether the new chemicals can match the performance of traditional materials without compromising quality. Consumer demand and stricter regulations, such as

Bridging the sustainability gap

- **Semiconductor giants lead the way.** Industry leaders like TSMC, Intel, and Samsung are driving sustainability efforts, cutting emissions, reducing energy consumption, and optimising materials. Their initiatives set industry standards, aiming for meaningful environmental impact.
- **Contrasts in electronics manufacturing.** While semiconductor leaders push sustainability, the broader electronics sector, particularly in Asia, shows mixed progress. Many manufacturers stick to cost-effective yet environmentally harmful processes. However, innovations like additive manufacturing for PCBs are emerging, offering a more sustainable alternative.
- **Opportunities in emerging markets.** India, the US, and Europe are seeing significant semiconductor investments fueled by policies like the Chips Act. New plants incorporate energy-efficient tech and compact cleanroom designs, embedding sustainability from the outset and avoiding the limitations of older facilities.
- **Legislation and cost efficiency.** Navigating regulations is crucial as governments tighten restrictions on toxic chemicals and mandate emissions reductions. The EU, Japan, and others enforce strict hazardous substance lists. Shifting to renewable energy and optimising energy and material use ensures compliance and cuts costs—simple measures like reducing rinse times and optimising standby modes boost both sustainability and profitability.
- **A call for industry-wide collaboration.** Sustainable manufacturing requires more than just new technology—it demands a fundamental shift in mindset and operations. As new facilities rise, especially in India, the industry has a chance to embed sustainability from the ground up, setting new benchmarks for the future.

the Eco-design for Sustainable Product Regulation in Europe and new global emissions standards, are crucial for adopting greener technologies. As these innovations progress, they help align electronics manufacturing with global sustainability goals.

Making electronics green

The shift towards sustainable materials in electronics manufacturing offers numerous benefits. Utilising biodegradable substrates, such as JIVA materials' flax-based substrates that dissolve in water, reduces environmental damage when products reach the end of their lifecycle. These substrates facilitate the retrieval of valuable metals like cadmium, gold, and silver from PCB components, which is especially useful in sectors such as electric vehicles for recycling precious metals from expired batteries.

However, incorporating these materials into traditional electronics manufacturing poses significant

challenges. The industry's reliance on established protocols complicates the introduction of new, potentially more expensive materials. Such transitions often require substantial initial investment and modifications to manufacturing processes and supply chains, necessitating extensive collaboration and testing.

Legislation plays a pivotal role in promoting sustainability. Regulatory frameworks mandating life cycle analyses are complex due to the intricate nature of electronic products and their supply chains. New regulations, such as Europe's digital product passport, which requires detailed emissions reporting and sustainability tracking, add layers of complexity to compliance.

AI tools are becoming crucial. Startups such as Banyan.eco use AI to help companies automate emissions data collection and sustainability tracking, enhancing the efficiency of life cycle assessments. This technology accelerates sustainability evaluations across the